MA2031- Linear Algebra for Engineers

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Module-5 (Lecture note-3)

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How to unitarily diagonalize a normal matrix?

Que: Suppose $A \in M_n(\mathbb{C})$ is normal. How to unitarily diagonalize A?

Suppose $A = UDU^*$ for some diagonal matrix D and unitary matrix U. Then $det(\lambda I - A) = det(U(\lambda I - D)U^*) = det(U)\ det(\lambda I - D)\ det(U^*) = det(\lambda I - D)$ for all $\lambda \in \mathbb{C}$. Hence

$$\sigma(A) = \sigma(D) = \text{ diagonal entries of D, say } \{d_1, \cdots, d_n\}.$$

Let $U = \big[u_1 | u_2 | \cdots | u_n\big] \in M_n(\mathbb{C})$, where $u_i \in \mathbb{C}^n$ be the columns of U. Then

$$\begin{aligned} A = UDU^* &\implies AU = UD \\ &\implies \big[Au_1 | Au_2 | \cdots | Au_n\big] = \big[d_1u_1 | d_2u_2 | \cdots | d_nu_n\big] \\ &\implies Au_j = d_ju_j \text{ for all } 1 \le j \le n. \end{aligned}$$

That is, $u_j \in \mathbb{C}^n$ is an eigenvector of A corresponding to the eigenvalue d_j.

Also U unitary implies that $\{u_1, u_2, \cdots, u_n\}$ is an ONB for $\mathbb{C}^n$. Thus,

- A unitarily diagonalizable $\implies \exists$ ONB for $\mathbb{C}^n$ consisting of eigenvectors of A.
- What about the converse?

Theorem (2A): Given a matrix $A \in M_n(\mathbb{C})$ the following are equivalent:

1. A is unitarily diagonalizable.
2. There exists an ONB for $\mathbb{C}^n$ consisting of eigenvectors of A.

Proof: $(1) \Rightarrow (2)$ Done !!
$(2) \Rightarrow (1)$ To find U and D such that $A = UDU^*$. By assumption there exists an ONB $\{u_i\}_{i=1}^n$ for $\mathbb{C}^n$ such that $Au_i = d_1 u_i$, where $\{d_1, d_2, \cdots, d_n\} = \sigma(A)$. Take

$$D = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}, \quad U = \begin{bmatrix} u_1 | u_2 | \cdots | u_n \end{bmatrix} \in M_n(\mathbb{C}).$$

Note that U is unitary and $AU = UD$, so that $A = UDU^*$.

Theorem (2B): Suppose $\mathcal{V}$ is a finite dimensional **complex IPS** and $T \in L(\mathcal{V})$.

Then the following conditions are equivalent:

1. There exists an orthonormal basis $\mathcal{B}$ such that ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is diagonal.
2. There exists an ONB for $\mathcal{V}$ consisting of eigenvectors of T.

Observation: Let $A = UDU^* \in M_7(\mathbb{C})$, where $U = [u_1|\cdots|u_7] \in M_7(\mathbb{C})$ and $D = diag\{d_1, \cdots, d_7\} \in M_7(\mathbb{C})$. Suppose

$$d_1 = d_3 \overset{(say)}{=:} \lambda_1; \quad d_2 = d_5 = d_7 \overset{(say)}{=:} \lambda_2; \quad d_4 \overset{(say)}{=:} \lambda_3; \quad d_6 \overset{(say)}{=:} \lambda_4.$$

Thus $\{\lambda_1, \cdots, \lambda_4\}$ are distinct eigenvalues of A. Let

$$\widetilde{U} = \left[u_1|u_3|u_2|u_5|u_7|u_4|u_6\right] \quad \text{and} \quad \widetilde{D} = diag\{\lambda_1, \lambda_1, \lambda_2, \lambda_2, \lambda_2, \lambda_3, \lambda_4\}.$$

Note that $\widetilde{U} \in M_7(\mathbb{C})$ is a unitary. Also

$$\begin{aligned} A\widetilde{U} &= \left[Au_1|Au_3|Au_2|Au_5|Au_7|Au_4|Au_6\right] \\ &= \left[d_1u_1|d_3u_3|d_2u_2|d_5u_5|d_7u_7|d_4u_4|d_6u_6\right] \\ &= \left[\lambda_1u_1|\lambda_1u_3|\lambda_2u_2|\lambda_2u_5|\lambda_2u_7|\lambda_3u_4|\lambda_4u_6\right] \\ &= \widetilde{U}\widetilde{D} \\ \implies \quad A &= \widetilde{U}\widetilde{D}\widetilde{U}^*. \end{aligned}$$

THUS: If A is normal, then there is unitary U and diagonal matrix D such that $A = UDU^*$ and the equal diagonal of D can be clubbed together.

Remark: Let $\mathcal{B}_i := \{u_1^{(i)}, \cdots, u_{n_i}^{(i)}\}$ be ONBs for $null(A - \lambda_i I)$, then

$\mathcal{B} = \cup_i \mathcal{B}_i$ is an ONB for $\mathbb{C}^n$ consisting of eigen values of A. Moreover,

$$U = \left[u_1^{(1)}|\cdots|u_{n_1}^{(1)}|u_1^{(2)}|\cdots|u_{n_2}^{(2)}|\cdots\cdots|u_1^{(r)}|\cdots|u_{n_r}^{(r)}\right] \in M_n(\mathbb{C})$$

is a unitary such that U^*AU is diagonal.

Example: Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, which is normal (in fact self-adjoint). Observe that

$\sigma(A) = \{0, 0, 3\} \overset{(say)}{=:} \{d_1, d_2, d_3\}$. Note that

$null(A - 0I) = \left\{(x_1, x_2, -x_2 - x_1)^t : x_i \in \mathbb{C}\right\} = \text{span}\{(0, 1, -1)^t, (1, 0, -1)^t\}$.

Apply Gram Schmidt process to get:

$$u_1 = \frac{1}{\sqrt{2}}(0, 1, -1)^t \quad \text{and} \quad u_2 = \frac{1}{\sqrt{6}}(2, -1, -1)^t.$$

Then $\{u_1, u_2\}$ is an ONB for $null(A - 0I)$. Since

$$null(A - 3I) = \{(x_1, x_1, x_1)^t : x_1 \in \mathbb{C}\} = \text{span}\{(1, 1, 1)^t\},$$

$u_3 := \frac{1}{\sqrt{3}}(1, 1, 1)^t$ is an ONB for $null(A - 3I)$. Thus $\{u_1, u_2, u_3\}$ is an eigenvector ONB for $\mathbb{C}^3$. Take

$$D = \begin{bmatrix} 0 & & \\ & 0 & \\ & & 3 \end{bmatrix} \quad \text{and} \quad U = \left[u_1 | u_2 | u_3\right] = \begin{bmatrix} 0 & \frac{2}{\sqrt{6}} & 1 \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & 1 \\ \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & 1 \end{bmatrix}$$

Then U is a unitary such that $A = UDU^*$ (verify).

Exercises

1. Which of the following matrices are unitarily diagonalizable. If yes, then find a unitary U which unitarily diagomalize it.

   (a) $\begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$. (b) $\begin{bmatrix} 1 & i & 0 \\ -i & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

2. Determine whether there exists an orthonormal basis $\mathcal{B}$ such that ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is diagona matrix, where T is given by:

   (a) $T : \mathbb{C}^2 \to \mathbb{C}^2$ defined by $T((x_1, x_2)^t) = (x_1 - ix_2, ix_1 + x_2)$.

   (b) $T : \mathbb{C}^3 \to \mathbb{C}^3$ by $T((x_1, x_2, x_3)^t) = (x_1 + ix_2, -ix_1 + x_2, 2x_3)$.

Observation: Suppose $A \in M_n(\mathbb{C})$ is normal. Then here exists unitary U such that

$$U^*AU = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}, \quad \text{where} \quad d_i \in \sigma(A).$$

Suppose $\{\lambda_1, \lambda_2, \cdots, \lambda_r\},\ r \leq n$, be distinct eigenvalues of A. By interchanging the columns of U (i.e, re-ordering the ONB) we assume that

$$U^*AU = \begin{bmatrix} D_1 & & \\ & \ddots & \\ & & D_r \end{bmatrix}, \quad \text{where} \quad D_j = \begin{bmatrix} \lambda_j & & \\ & \ddots & \\ & & \lambda_j \end{bmatrix} = \lambda_j I_{n_j} \in M_{n_j}(\mathbb{C}), 1 \leq \ n_j \leq n.$$

$$\implies U^*AU = \begin{bmatrix} D_1 & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} + \begin{bmatrix} 0 & & & \\ & D_2 & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} + \cdots + \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & \ddots & \\ & & & D_r \end{bmatrix}$$

$$\implies U^*AU = \lambda_1 \begin{bmatrix} I_{n_1} & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 & & & \\ & I_{n_2} & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} + \cdots + \lambda_r \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & \ddots & \\ & & & I_{n_r} \end{bmatrix}$$

$$\Longrightarrow A = \lambda_1 U \begin{bmatrix} I_{n_1} & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} U^* + \lambda_2 U \begin{bmatrix} 0 & & & \\ & I_{n_2} & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} U^* + \cdots + \lambda_r U \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & \ddots & \\ & & & I_{n_r} \end{bmatrix} U^*$$

$$\overset{(say)}{=} \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_r P_r$$

Note that $P_j = P_j^* = P_j^2$ (i.e., orthogonal projection) and $P_j P_k = 0 \ (j \neq k)$. Also

$$\sum_j P_j = U\left(\begin{bmatrix} I_{n_1} & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} + \begin{bmatrix} 0 & & & \\ & I_{n_2} & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} + \cdots + \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & \ddots & \\ & & & I_{n_r} \end{bmatrix} \right) U^* = UU^* = I_n.$$

THUS: A normal implies $A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_r P_r$ where λ_i's are distinct eigenvalues of A, and P_j's are orthogonal projections with $P_j P_k = 0, \ (j \neq k)$ and $\sum_j P_j = I$.

Question: What about the converse?

Theorem (3A): Given a matrix $A \in M_k(\mathbb{C})$ the following are equivalent:

1. A is normal.
2. If $\{\lambda_1, \cdots, \lambda_r\}$ are the distinct eigen values of A, then there exists orthogonal projections $P_1, \cdots, P_r$ with $P_j P_k = 0$ $(j \neq k)$ and $\sum_j P_j = I$ such that $A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_r P_r$.

Proof $(1) \Rightarrow (2)$ Done !! TO prove $(2) \Rightarrow (1)$ consider

$$\begin{aligned} A^*A &= (\lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_r P_r)^*(\lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_r P_r) \\ &= \sum_j \sum_k \overline{\lambda_j} \lambda_k P_j P_k = \sum_j \overline{\lambda_j} \lambda_j P_j = \sum_j |\lambda_j|^2 P_j. \end{aligned}$$

Similarly, $AA^* = \sum_j |\lambda_j|^2 P_j$. Thus A is normal. $\square$

Theorem (3B): Suppose $\mathcal{V}$ is a finite dimensional **complex IPS** and $T \in L(\mathcal{V})$.

Then the following conditions are equivalent:

1. T is normal.
2. If $\{\lambda_1, \cdots, \lambda_r\}$ are the distinct eigen values of T, then there exists orthogonal projections $P_1, \cdots, P_r \in L(\mathcal{V})$ with $P_j P_k = 0$ $(j \neq k)$ and $\sum_j P_j = I$ such that $T = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_r P_r$.

Spectral theorem for normal matrices

<u>**Theorem:**</u> Let $A \in M_n(\mathbb{C})$ and $\{\lambda_1, \cdots, \lambda_r\}$ be the distinct eigenvalues of A.

Then the following conditions are equivalent:

1. A is normal.
2. A is unitarily diagonalizable.
3. There exists an ONB for $\mathbb{C}^n$ consisting of eigenvectors of A.
4. There exists orthogonal projections $P_1, \cdots, P_r \in M_n(\mathbb{C})$ with $P_j P_k = 0$ $(j \neq k)$ and $\sum_j P_j = I$ such that $A = \lambda_1 P_1 + \cdots\cdots + \lambda_r P_r$.
5. $\mathbb{C}^n = null(A - \lambda_1 I) \overset{\perp}{\oplus} null(A - \lambda_2 I) \overset{\perp}{\oplus} \cdots\cdots \overset{\perp}{\oplus} null(A - \lambda_r I)$.

Proof: $(1) \Leftrightarrow (2) \Leftrightarrow (3) \Leftrightarrow (4)$ Done !! $(3) \Leftrightarrow (5)$ Exercise. $\square$

Spectral theorem for normal operators

Theorem: Let $\mathcal{V}$ be finite-dimensional **complex** inner-product space and $T : \mathcal{V} \to \mathcal{V}$ be a linear map. Then the following are equivalent:

1. T is normal.
2. There exists an ONB, say $\mathcal{B}$, for $\mathcal{V}$ such that $[T]_{\mathcal{B}}$ is diagonal.
3. There exists an ONB for $\mathcal{V}$ consisting of eigenvectors of T.
4. If $\{\lambda_1, \cdots, \lambda_r\}$ are the distinct eigenvalues of T, then there exists orthogonal projections $P_1, \cdots, P_r$ with $P_j P_k = 0$ $(j \neq k)$ and $\sum_j P_j = I$ such that
$$T = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_r P_r.$$
5. If $\{\lambda_1, \cdots, \lambda_r\}$ are the distinct eigen values of T, then
$$\mathcal{V} = null(T - \lambda_1 I) \overset{\perp}{\oplus} null(T - \lambda_2 I) \overset{\perp}{\oplus} \cdots \cdots \overset{\perp}{\oplus} null(T - \lambda_r I).$$

Proof: Exercise. □

Example: Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, which is normal (in fact self-adjoint). Observe that

$\sigma(A) = \{0, 0, 3\} \overset{(say)}{=:} \{d_1, d_2, d_3\}$. Note that

$null(A - 0I) = \left\{(x_1, x_2, -x_2 - x_1)^t : x_i \in \mathbb{C}\right\} =$ span$\{(0, 1, -1)^t, (1, 0, -1)^t\}$.

Apply Gram Schmidt process to get:

$$u_1 = \frac{1}{\sqrt{2}}(0, 1, -1)^t \quad \text{and} \quad u_2 = \frac{1}{\sqrt{6}}(2, -1, -1)^t.$$

Then $\{u_1, u_2\}$ is an ONB for $null(A - 0I)$. Since

$$null(A - 3I) = \{(x_1, x_1, x_1)^t : x_1 \in \mathbb{C}\} = \text{span}\{(1, 1, 1)^t\},$$

$u_3 := \frac{1}{\sqrt{3}}(1, 1, 1)^t$ is an ONB for $null(A - 3I)$. Thus $\{u_1, u_2, u_3\}$ is an eigenvector ONB for $\mathbb{C}^3$. If $D =$diag$\{0, 0, 3\}$ and $U = \left[u_1 | u_2 | u_3\right]$, then $A = UDU^*$ (verify).
Let

$$P_1 := \frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \quad \text{and} \quad P_2 := \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Note that $P_j^2 = P_j = P_j, P_1 P_2 = P_2 P_1 = 0$ and $P_1 + P_2 = 1$.
Further $A = 0P_1 + 3P_2$.

Example: Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \in M_3(\mathbb{C})$. Note that $A^*A \neq AA^*$, hence A is not unitarily diagonalizable. The characteristic polynomial is $p_A(\lambda) = -(\lambda-1)^2(\lambda-2)$, so $\sigma(A) = \{1, 1, 2\}$. Note that

$$null(A - I) = \{(x_1, 0, 0)^t : x_1 \in \mathbb{C}\} = \text{span}\{(1, 0, 0)^t\}$$
$$null(A - 2I) = \{(0, 0, x_3)^t : x_3 \in \mathbb{C}\} = \text{span}\{(0, 0, 1)^t\}$$

So there do not exist three linearly independent eigenvectors of A, hence A is not "diagonalizable".

Example: Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \in M_2(\mathbb{C})$. Note that $A^*A \neq AA^*$, hence A is not unitarily diagonalizable. Note that $\sigma(A) = \{1, 2\}$, distinct eigenvalues, hence "diagonalizable". Clearly

$$null(A - I) = \text{span}\{(1, 0)^t\} \quad \text{and} \quad null(A - 2I) = \text{span}\{(1, 1)^t\}.$$

Observe that $\mathbb{C}^2 = null(A - I) \oplus null(A - 2I)$, but is not a orthogonal direct sum. So there is no ONB for $\mathbb{C}^2$ consisting eigenvectors of A.